

Floor planning challenges

Hierarchy placement. The design is physically flat but the logical hierarchy is maintained. The automatic macro placer identified these top-level hierarchies and the area they needed to be placed. Results of the placer are taken as reference to further the macro placement. Initially, while placing the memories, the individual hierarchy is picked up. Then the internal hierarchies are highlighted and memories with the same hierarchy placed in a stack.

Stack preparation. Stacks of memories are created as the design contains 640 memories. Memories are of vivid aspect ratio and different pin densities, which increased the complexity in stacking. One of the challenges in stacking is that smaller memories have pin counts similar to bigger ones. To tackle this hurdle, smaller memories are kept close to the core area in order to provide them better routing accessibility and reduce congestion.

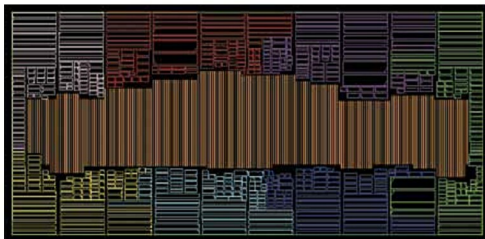


Fig. 2: Special mesh structure in power plan

Channel-width estimation. The main concern while doing this is to identify the approximate channel widths for both the two memories in a stack, and vertical channel width between two stacks of memories. The width of the horizontal channels increases gradually as the channel buffering of any memory stack increases from the block boundary to the block core area.

Pyramidal memory stacking. It's a special methodology developed to overcome congestion issue. The vertical channel requirement at boundary is lesser than near the core interface because the routing resources and buffering required near the core area is more. This methodology helps tackle this issue by creating a vertical channel in a pyramidal shape from the core to the boundary.



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